AI-Based Sea Level Prediction to Support Long Island Communities

— Using Machine Learning to Support Coastal Resilience —

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Presidential AI Challenge Submission

Sea Level is rising

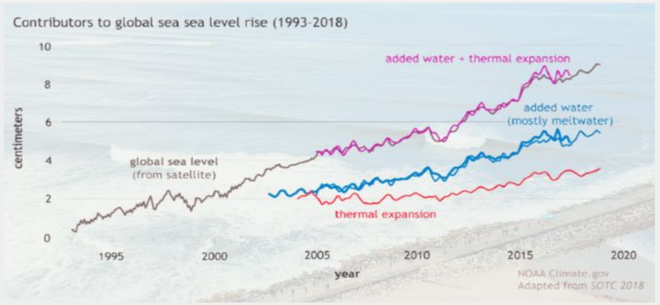

Fig 1: Contributors to global sea level rise since 1993 (NOAA Climate.gov, 2019).

- The speed of sea level rise has doubled in the last century (Oppenheimer et al., 2019).
- Sea level rise increased coastal flooding risk (Church et al., 2013).
- Data driven models help improve prediction (Ahmed et al., 2021).

What is causing sea level to rise?

- **Driving Factors**
  - Temperature
  - AMOC
  - El Niño
  - Greenland Ice
  - Antarctica Ice
  - CO_2

Fig 2: Factors on Long Island Sea Level Rise (Student Researchers, 2026).

Vulnerability of Long Island

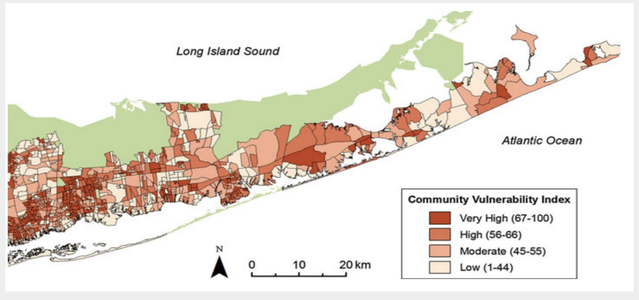

Fig 3: Community Vulnerability Index of Southern Suffolk County *(Shepard et al., 2012).*

Sea level rising increases the frequency and severity of coastal flooding, accelerates shoreline erosion, and drives saltwater intrusion into freshwater aquifers, threatening ecosystems, infrastructure and human settlements and is a particularly important factor to include when assessing future hazard risks, especially for the densely populated coastal areas (Nicholls et al., 2007) (Church et al., 2013).

Problem and Need

- ➢ Rising sea levels increase flooding risk on Long Island
- ➢ Regional sea level rise exceeds global average
- ➢ Tens of thousands of homes at risk by 2040–2070
- ➢ Communities need localized, accurate predictions

Project Goal

- ➢ Develop an AI-based model to predict future sea level changes
- ➢ Along the Long Island coast to support planning and resilience

Why Artificial Intelligence?

- ➢ Captures nonlinear climate–ocean interactions
- ➢ Faster predictions than traditional models
- ➢ Improved local-scale accuracy
- ➢ Enables scenario-based forecasting

Community Impact

- ➢ Supports coastal planning and resilience
- ➢ Improves flood risk awareness
- ➢ Assists policymakers and emergency managers
- ➢ Scalable to other coastal regions

The Importance of Greenland

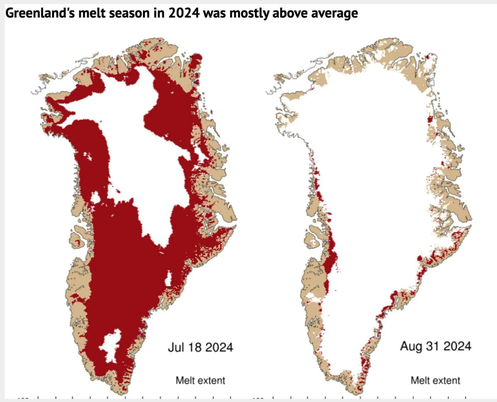

Fig 4: Map showing areas of Greenland undergoing surface melt on 18 July and 31 August 2024 (shaded red) (Martin Stendel et al., 2024).

The Greenland ice sheet is a growing contributor to global sea-level rise. Recent studies on satellite observations reveal that the magnitude of recent GrIS melting has been exponential over the last 350 years, which is dominated by surface meltwater runoff (Trusel et al., 2018).

AI Models

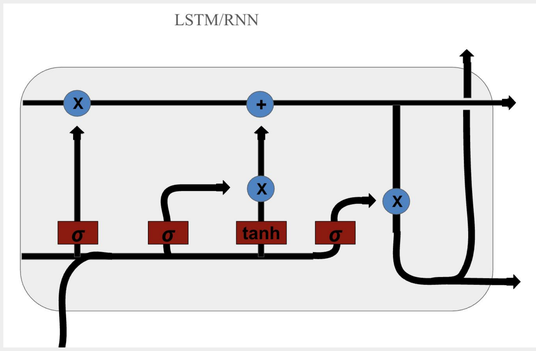

Fig 5: Architecture model of LSTM/RNN model (Student Researchers, 2025). The model processes sequential input data over time, allowing formation from previous time steps to influence current predictions. LSTM units will include memory cells that keep long-term information to improve prediction accuracy for time dependent sea level data.

Six models will be used

- Linear Regression
- Random Forest
- XGBoost
- Gaussian Process
- RNN
- LSTM

Training Data Sources

Factor	Unit	Source	Dataset
Historical Sea Level	mm	NOAA Tide and Currents	Verified Monthly Mean Sea level
Temperature*	Celsius	NOAA National Centers for Environmental Information	NOAAGlobalTemp
Greenland Ice Loss	Gigatons	NASA Goddard Space Flight Center	GRACE / GRACE-FO Ice Mass Change (Greenland)
Antarctic Ice Loss	Gigatons	NASA Goddard Space Flight Center	GRACE / GRACE-FO Ice Mass Change (Antarctica)
AMOC	Sverdrups	Met Office Climate Dashboard / RAPID Array	RAPID-MOCHA AMOC Time Series
ENSO	Sigma	NOAA Climate Prediction Center	Niño 3.4 Index
CO_2	ppm	NOAA GLobal Monitoring Laboratory	Mauna Loa CO_2 Monthly Mean